Congestion and heterogenous quadratic regularization in optimal transport

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Abstract

In this paper, we build upon the optimal transport quadratic regularization model to develop a framework that incorporates congestion costs, particularly in matching within the healthcare and education sectors. Specifically, we introduce a model with heterogeneous quadratic costs. We analyze the model's properties under specific cases, extending the existing literature. Furthermore, we explore key structural characteristics of the model and provide numerical examples illustrating why this formulation more accurately captures real-world phenomena, particularly in the Peruvian context. The main result consists of identifying a specific type of corner solution when matching the same number of clusters, i.e., $N = L$.

Keywords: optimal transport, congestion costs, quadratic regularization, matching.
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1 Introduction

Matching theory in economics studies how agents are paired based on their preferences and market constraints. The seminal work of Gale and Shapley (1962) introduced the concept of stable matching, where no two agents prefer each other over their assigned partners. This theory was later extended by Hylland and Zeckhauser (1979) in the context of house allocation and by Kelso and Crawford (1982), who incorporated transfers in two-sided matching. Alvin Roth significantly advanced these ideas, applying them to real-world scenarios such as school admissions and organ transplants, demonstrating the effectiveness of matching mechanisms (Roth (1982), Roth and Sotomayor (1990)). He also proved that no stable matching mechanism ensures truthfulness as a dominant strategy (Roth (1982)). In school choice, Abdulkadiroğlu and Sönmez (2003) developed mechanisms that balance stability and individual preferences. Further contributions include Hatfield and Milgrom (2005), Echenique and Yenmez (2015), and the recent book Echenique et al. (2023), which provides a comprehensive overview of these subjects.

In recent years, matching theory has been enriched by Optimal Transport (OT) methods, a mathematical framework introduced by Monge in the 18th century and rigorously formalized by Kantorovich in the 20th century. OT addresses optimal assignment problems by minimizing transportation costs and has been widely applied in matching markets, including student-school, patient-hospital, and worker-firm pairings. The foundational book by Cédric Villani (Villani (2009)) provides an exhaustive treatment of OT, while Ekeland (2010) and Ambrosio et al. (2021) further develop its mathematical aspects. In economics, Alfred Galichon bridged OT with discrete matching problems in markets (Galichon (2016), Galichon (2021)). The main advantage of OT is its ability to model agents as continuous distributions rather than discrete entities, optimizing over distributions. A recent study, Echenique et al. (2024), explores stability in matching through an OT framework applicable to both discrete and continuous settings. They show that utility transformations parameterized by α yield ε-stable, welfare-maximizing, or ε-egalitarian solutions, depending on whether the transformation is convex or concave. This approach aligns with Niederle and Yariv (2009) and Ferdowsian et al. (2023). In a discrete setting, Galichon (2021) employs computational methods (SISTA algorithm) akin to Merigot and Thibert (2020) and Nenna (2020) to recover matching costs in migration problems, with further applications in marriage and labor markets (Dupuy and Galichon (2014), Dupuy et al. (2019)).

In this work, we develop a variant of the OT model with quadratic heterogenous regularization. The model captures congestion costs and provides new insights relative to the existing literature. In particular, after addressing the problem from a theoretical perspective using basic elements of convex nonlinear programming, we present examples that illustrate our approach. The remainder of the paper is organized as follows. In Section 2, we define the notation and present existing models from the literature. Next, in Section 3, we introduce our model. We then derive some new results and conclude with examples and extensions. These examples account for the economic insights brought by our model.

2 Preliminaries

We consider two sets, $X = \{x_1, \ldots, x_N\}$ and $Y = \{y_1, \ldots, y_L\}$. Each element $x_i \in X$ represents a group/type of students or patients, and each $y_j \in Y$ represents a school or hospital[1]. To each x_i (y_j), we associate a *mass* μ_i (ν_j) in $\mathbb{Z}_{++}$, corresponding to the number of individuals in the group (school capacity or hospital bed availability). For simplicity, we refer to elements of X by their index i and elements of Y by their index j. We denote by π_{ij} the number of individuals of type i assigned to entity j (e.g., students to schools or patients to hospitals).

The problem, from a central planner's perspective, is to decide how many individuals from group i should be assigned to $j = 1, \ldots, L$, while minimizing the matching cost, given by a function $C : \mathbb{R}_+^{N,L} \times \mathbb{R}^P \to \mathbb{R}$, depending on the matching $\pi = [\pi_{ij}] \in \mathbb{R}_+^{N,L}$[2], and a vector of parameters $\theta \in \mathbb{R}^P$. The central planner's objective is to minimize this cost[3], subject to the constraints:

$$\Pi(\mu, \nu) = \left\{ \pi_{ij} \geq 0 : \sum_{j=1}^{L} \pi_{ij} = \mu_i, \ \forall\, i \in I \ \wedge \ \sum_{i=1}^{N} \pi_{ij} = \nu_j, \ \forall\, j \in J \right\}. \tag{1}$$

Constraints (1) ensure that all individuals (students, patients, etc.) are assigned and that all entities (schools, hospitals, etc.) fill their available capacity. Thus, the central planner solves:

$$\min_{\pi \in \Pi(\mu,\nu)} C(\pi; \theta). \tag{2}$$

The planner must choose an optimal matching[4] π that minimizes the cost function. A solution to (2) is referred to as an optimal matching or optimal (transport) plan, denoted by π^*.

In the literature, the standard optimal transport model assumes separable linear costs, i.e., $C(\pi, \theta) = \sum_{i,j} c_{ij}\pi_{ij}$. This implies that the marginal cost of assigning an additional individual from i to j is constant, regardless of the existing assignments. Hence, the planner solves:

$$\mathcal{P}_O : \quad \min_{\pi \in \Pi(\mu,\nu)} \sum_{i,j} c_{ij}\pi_{ij}.$$

To solve $\mathcal{P}_O$, linear programming techniques such as the simplex method are typically used.

The classical OT model extends to continuous settings, optimizing over distributions rather than discrete entities. However, in this work, we remain in a discrete framework, inspired by the entropic an quadratic regularization approaches (Carlier et al., 2020; Peyré and Cuturi, 2019;

[1] In this work we keep in mind these contexts. However, this model can be extended to other situations: labor market, migration etc.

[2] In this work, we mostly assume that the number of individuals matched can take real positive values, not just positive integers. This assumption is analogous to utility maximization problems where goods are treated as divisible.

[3] Matching individuals involves costs beyond «physical» transportation, including implicit factors such as tuition fees, admission criteria, medical specialties, insurance coverage, and personal preferences. Thus, we refer to these as matching costs rather than transportation costs.

[4] In some cases, π is treated as a vector instead of a matrix: $\pi = (\pi_{11}, \pi_{12}, \cdots, \pi_{NL})^T$.

Nutz, 2024). With respect to entropic regularization, the problem takes the form:

$$\min_{\pi \in \Pi(\mu,\nu)} \sum_{i=1}^{N} \sum_{j=1}^{L} c_{ij}\pi_{ij} + \sigma \pi_{ij} \ln(\pi_{ij}),$$

with $\sigma > 0$. On the other hand, quadratic regularization (Wiesel and Xu, 2024; Nutz, 2024) changes the entropy by $(\varepsilon/2)||\pi||_2^2$. This is the most similar formulation already existing to our model.

Returning to the discrete setting with linear costs, if π is taken in $\mathbb{Z}_+^{N,L}$, the central planner's problem always has a solution since there is a finite number of possible assignments.

Proposition 2.1. In an integer setting, the number of matchings is at most L^M.

Proof. The number of ways to assign all μ_i individuals from group i to entities is given by solutions to:

$$\pi_{i1} + \cdots + \pi_{iL} = \mu_i, \quad 0 \leq \pi_{ij} \leq \nu_j \quad \forall\, j = 1, \cdots, L. \tag{3}$$

Ignoring the upper bounds ν_j, this reduces to a stars and bars problem. The upper bound for the number of solutions to (3) is $\binom{\mu_i+L-1}{L-1}$. Applying the multiplication principle, the total number of matchings satisfies:

$$\prod_{i=1}^{N} \binom{\mu_i + L - 1}{L-1} = \prod_{i=1}^{N} \prod_{j=1}^{\mu_i} \frac{j+L-1}{j} \leq \prod_{i=1}^{N} \prod_{j=1}^{\mu_i} L = L^M. \qquad \blacksquare$$

Proposition 2.1 guarantees the existence of a solution in the integer setting. However, $\mathcal{P}_O$ does not enforce $\pi_{ij} \in \mathbb{Z}_+$, potentially allowing non-integer solutions. Despite this, Weierstrass Theorem ensures that $\mathcal{P}_O$ always has a solution in $\mathbb{R}_+^{N,L}$.

Proposition 2.2. Given $\mu = (\mu_1, ..., \mu_N)^T \in \mathbb{R}_{++}^N$ and $\nu = (\nu_1, ..., \nu_L)^T \in \mathbb{R}_{++}^L$, $\mathcal{P}_O$ always has a solution π^*.

Proof. The objective function is continuous as it is linear. The constraint set $\Pi(\mu, \nu)$ is compact in $\mathbb{R}^{NL}$ since it is the intersection of closed sets and bounded within $[0, M]^{NL}$. $\blacksquare$

The basic model has been extensively studied, with variations such as entropic (Dupuy and Galichon, 2014; Carlier et al., 2020) or quadratic (Lorenz et al., 2019; González-Sanz and Nutz, 2024; Wiesel and Xu, 2024; Nutz, 2024) regularization, or in the continuous framework (Dupuy and Galichon, 2014; Echenique et al., 2024). To the best of our knowledge, the model and results we deliver are new in the literature.

3 Congestion costs

We now introduce a new variant of the optimal transport problem in the discrete setting that explicitly accounts for congestion effects. Traffic congestion and institutional overload are crucial factors affecting the allocation of individuals to entities such as schools and hospitals.

When too many individuals are matched to the same entity, congestion costs escalate, leading to inefficiencies in both physical and bureaucratic dimensions. This phenomenon is observed in various settings:

- **Traffic congestion:** The simultaneous assignment of many students to the same school in urban areas can increase travel times, overload public transport, and generate bottlenecks in key traffic zones. The same happens with patients and hospitals.
- **Medical centers overload:** Large patient inflows can overwhelm hospital resources, creating long waiting times, administrative bottlenecks, and inefficient service delivery.
- **Bureaucratic congestion:** Excess demand for certain institutions may slow down processing times, affecting school admissions, hospital triage, and public service allocation due to outdated systems and inefficient workflows.

To model this phenomenon, we consider a strictly convex cost function with respect to the number of matched individuals, capturing the increasing marginal costs associated with congestion.

3.1 Mathematical Formulation

We define the cost function $C(\pi;\theta)$[5] as a separable function:

$$C(\pi;\theta) = \sum_{i=1}^{N}\sum_{j=1}^{L} \phi(\pi_{ij};\theta_{ij}), \tag{4}$$

where ϕ_{ij} is structurally homogeneous[6]. The central planner's problem then becomes:

$$\min_{\pi\in\Pi(\mu,\nu)} \left\{ \sum_{i=1}^{N}\sum_{j=1}^{L} \phi(\pi_{ij};\theta_{ij}) \right\}, \tag{5}$$

where $\Pi(\mu,\nu)$ is defined as in (1). Given that congestion leads to increasing costs, ϕ should be strictly increasing and strictly convex, transforming the problem into a convex optimization problem with linear constraints.

Although the Linear Independence Constraint Qualification (LICQ) condition may fail for solutions where non-negativity constraints are not binding, the convexity of the objective function and the linearity of constraints allow us to apply the Karush-Kuhn-Tucker (KKT) conditions, see Boyd (2004).

[5] As mentioned earlier, π can be taken as a vector in $\mathbb{R}_+^{NL}$ rather than a matrix in $\mathbb{R}_+^{N,L}$.

[6] The function φ_{ij} does not change structurally across (i,j) pairs; whether logarithmic, exponential, or polynomial, we assume $\varphi_{ij} = \varphi$.

3.2 Lagrangian Formulation and KKT Conditions

The Lagrangian function associated with (5) is given by:

$$\mathscr{L}(\pi,\lambda,\xi,\gamma;\theta) = \sum_{i=1}^{N}\sum_{j=1}^{L}\phi(\pi_{ij};\theta_{ij}) + \sum_{i=1}^{N}\xi_i\left(\mu_i - \sum_{j=1}^{L}\pi_{ij}\right) + \sum_{j=1}^{L}\lambda_j\left(\nu_j - \sum_{i=1}^{N}\pi_{ij}\right) - \sum_{i=1}^{N}\sum_{j=1}^{L}\gamma_{ij}\pi_{ij}. \tag{6}$$

The KKT first-order conditions are:

$$\begin{aligned}
\frac{\partial\mathscr{L}(\pi^*,\xi^*,\lambda^*,\gamma^*;\theta)}{\partial\pi_{ij}} &= \frac{\partial\phi(\pi^*_{ij};\theta_{ij})}{\partial\pi_{ij}} - \lambda^*_j - \xi^*_i - \gamma^*_{ij} = 0,\ \forall\ (i,j)\in I\times J,\\
-\pi^*_{ij} &\le 0,\ \forall\ (i,j)\in I\times J,\\
\sum_{j=1}^{N}\pi^*_{ij} - \mu_i &= 0,\ \forall\ i\in I,\\
\sum_{i=1}^{N}\pi^*_{ij} - \nu_j &= 0,\ \forall\ j\in J,\\
\gamma^*_{ij}\pi^*_{ij} &= 0.
\end{aligned}$$

Since the objective function is strictly convex and the constraint set is convex, the solution is unique.

3.3 Quadratic Cost Function and Optimal Matching

To carry out a quantitative analysis, we assume a quadratic cost function:

$$\phi(\pi_{ij};\theta_{ij}) = d_{ij} + c_{ij}\pi_{ij} + a_{ij}\pi^2_{ij}. \tag{7}$$

Thus, the optimization problem becomes:

$$\mathcal{P}_1 :\ \min_{\pi\in\Pi(\mu,\nu)}\left\{\sum_{i=1}^{N}\sum_{j=1}^{L} d_{ij} + c_{ij}\pi_{ij} + a_{ij}\pi^2_{ij}\right\}. \tag{8}$$

The parameters have clear economic interpretations:

- d_{ij} represents fixed costs associated with each matching (e.g., baseline administrative or transportation costs).
- c_{ij} corresponds to constant marginal costs, capturing individual and pair characteristics.
- a_{ij} introduces congestion effects, ensuring increasing marginal costs as π_{ij} grows.

Applying the KKT conditions, we obtain:

$$\pi^*_{ij} = \frac{\xi^*_i + \lambda^*_j + \gamma^*_{ij} - c_{ij}}{2a_{ij}}. \tag{9}$$

3.4 Structural Properties of the Solution

A fundamental issue in this model is determining whether solutions are interior ($\pi^*_{ij} > 0$ for all (i,j)) or a corner solutions ($\gamma^*_{ij} > 0$ for some (i,j)). The following result characterizes a structural property of the solution:

Proposition 3.1. With respect to the problem (5), with costs given by (7), whenever $\gamma^*_{ij} = 0$ for all $(i,j) \in I \times J$, the linear system obtained from (9), with respect to (ξ^*, λ^*), leads to a singular $N + L$ linear system.

Proof. Since $\gamma^*_{ij} = 0$ for all $(i,j) \in I \times J$, first order conditions lead to

$$\sum_{j=1}^{L} \pi^*_{ij} = \sum_{j=1}^{L} \frac{\xi^*_i}{2a_{ij}} + \sum_{j=1}^{L} \frac{\lambda^*_j}{2a_{ij}} - \sum_{j=1}^{L} \frac{c_{ij}}{2a_{ij}} = \mu_i, \ \forall\, i \in I \tag{10}$$

$$\sum_{i=1}^{N} \pi^*_{ij} = \sum_{i=1}^{N} \frac{\xi^*_i}{2a_{ij}} + \sum_{i=1}^{N} \frac{\lambda^*_j}{2a_{ij}} - \sum_{i=1}^{N} \frac{c_{ij}}{2a_{ij}} = \nu_j, \ \forall\, j \in J. \tag{11}$$

By setting $x = \begin{bmatrix} \xi^*_1 & \cdots & \xi^*_N & \lambda^*_1 & \cdots & \lambda^*_L \end{bmatrix}^T \in \mathbb{R}^{N+L}$, the linear equalities (10) and (11) on ξ^*_i and λ^*_j are described by the linear system $(\Lambda + T)x = b$, where

$$\Lambda = \mathrm{Diag}\left(\sum_{j=1}^{L} \frac{1}{2a_{1j}}, \ldots, \sum_{j=1}^{L} \frac{1}{2a_{Nj}}, \sum_{i=1}^{N} \frac{1}{2a_{i1}}, \ldots, \sum_{i=1}^{N} \frac{1}{2a_{iL}} \right) \in \mathbb{R}^{N+L,N+L}.$$

$$\Upsilon = \left[\frac{1}{2a_{ij}} \right]_{\substack{1 \le i \le N \\ 1 \le j \le L}} \in \mathbb{R}^{N,L} \text{ and } T = \begin{bmatrix} 0 & \Upsilon \\ \Upsilon^T & 0 \end{bmatrix} \in \mathbb{R}^{N+L,N+L},$$

$$b = \left[\mu_1 + \sum_{j=1}^{L} \frac{c_{1j}}{2a_{1j}}, \ldots, \mu_N + \sum_{j=1}^{L} \frac{c_{Nj}}{2a_{Nj}}, \nu_1 + \sum_{i=1}^{N} \frac{c_{i1}}{2a_{i1}}, \ldots, \nu_L + \sum_{i=1}^{N} \frac{c_{iL}}{2a_{iL}} \right]^T \in \mathbb{R}^{N+L}.$$

Let $R = \Lambda + T$. If R_k denotes the $k-$th row of R, we note that $R_1 = \sum_{k=N+1}^{N+L} R_k - \sum_{k=2}^{N} R_k$. Hence, $\mathrm{Det}(R) = 0$, and the claim follows. ■

As usual in economics, we are interested in perform monotone or smooth comparative statics. With respect to the former (see Milgrom and Shannon (1994)), it can't be performed since $S = \Pi(\mu, \nu)$ is not a sub-lattice of $X = \mathbb{R}^{NL}_+$. Indeed, given $\pi_1, \pi_2 \in S$, in general, $\pi_1 \wedge \pi_2$ and $\pi_1 \vee \pi_2$ do not belong to S. With respect to the latter, Proposition 3.2 explains why smooth comparative statics cannot be accomplished.

Proposition 3.2. With respect to (6), considering quadratic costs[7]

$$\mathrm{Det}(J_{\pi,(\xi,\lambda)} \overline{\mathscr{L}}(\pi^*, \xi^*, \lambda^*, \overline{\theta})) = 0.$$

[7] Following de la Fuente (2000) notation. Here $\overline{\mathscr{L}} = (\nabla_\pi \mathscr{L}, \nabla_\theta \mathscr{L})$.

Proof. First, let $\pi = (\pi_{11}, \ldots, \pi_{1L}, \cdots, \pi_{N1}, \ldots, \pi_{NL})^T$. Then, we define

$$D = \mathrm{Diag}(a_{11}, \ldots, a_{1L}, \cdots, a_{N1}, \ldots, a_{NL}) \in \mathbb{R}_{++}^{NL,NL}$$

and $B = [b_{k\ell}] \in \mathbb{R}^{N+L,N+L}$, where

$$b_{k\ell} = \begin{cases} 1 & \text{if } k \leq N \text{ and } (k-1)L < \ell \leq kL, \\ 1 & \text{if } N < k \leq N+L \text{ and } \ell \equiv k - N \pmod{L}, \\ 0 & \text{otherwise.} \end{cases}$$

Matrix B never has full rank. Indeed, $B_1 = \sum_{k=N+1}^{N+L} B_k - \sum_{k=2}^{N} B_k$, where B_k is row k of B. Thus, since

$$J_{\pi,(\xi,\lambda)}\overline{\mathscr{L}}(\pi^*, \xi^*, \lambda^*, \overline{\theta}) = \begin{bmatrix} D & -B^T \\ -B & 0 \end{bmatrix},$$

following Gentle (2017), $\mathrm{Det}(J_{\pi,(\xi,\lambda)}\overline{\mathscr{L}}(\pi^*, \xi^*, \lambda^*, \overline{\theta})) = \mathrm{Det}(D)\mathrm{Det}(0 - BD^{-1}B^T) = 0$. ■

Although we cannot apply smooth comparative statics, the conditions of the Envelope Theorem are satisfied for π^* in the interior of Π. Therefore, by defining $V = V(\pi^*) = \sum_{i=1}^{N}\sum_{j=1}^{L} \phi_{ij}(\pi_{ij}^*; \overline{\theta}_{ij})$, we can conclude that $\partial V/\partial c_{ij} = \pi_{ij}^* > 0$ and $\partial V/\partial a_{ij} = \pi_{ij}^{*2} > 0$, which is expected, as the cost of the optimal transport plan only increases if the coefficients associated with preference costs and congestion costs rise.

Note that, in general, obtaining the optimal matching π^* from (9), is quite complicated. Even if we assume an interior solution, which would simplify the equations since $\gamma_{ij}^* = 0$ automatically, we still cannot solve the linear system systematically. Note also that R not being invertible does not imply that the system has no solution. It only means that, if a solution (ξ^*, λ^*) exists, it is either not unique, or there is $\gamma_{ij}^* \neq 0$. What is unique is π^* since the objective function is strictly convex. Hence, even if we have several (ξ^*, λ^*), at the end, we obtain a unique π^*. The non uniqueness of (ξ^*, λ^*) originates from the fact that the LICQ does not hold for interior solutions.

Nonetheless, when $N = L$, we do can obtain a explicit solution for our model under mild assumptions.

4 Analysis for $N = L$

An important issue in our model is to determine whether the solution will be a corner solution ($\gamma_{ij}^* > 0$ for some $(i, j) \in I \times J$) or not. The following examples show that under the quadratic setting, both interior and corder solutions can exist.

Example 4.1. In this example, we show a case where the solution is interior. Consider

$$a = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}, \; c = \begin{bmatrix} 24 & 48 \\ 16 & 24 \end{bmatrix}, \; \mu = (20, 20), \; \nu = (12, 28), \; \text{so that } N = L = 2.$$

Then, we have $\pi^* = (7, 13, 5, 15)$.

Example 4.2. To illustrate a case where the solution is a corner solution, consider the following values:

$$a = \begin{bmatrix} 200 & 2 \\ 2 & 200 \end{bmatrix}, \quad c = \begin{bmatrix} 200 & 2 \\ 2 & 200 \end{bmatrix}, \quad \mu = (10, 10), \quad \nu = (10, 10), \quad \text{so that } N = L = 2.$$

In this scenario, the optimal solution is $\pi^* = (0, 10, 10, 0)$, a corner solution.

Note that in Example 4.1, the solution no longer satisfies the property of the classical model where there always exists $(i, j) \in I \times J$ such that $\pi^*_{ij} = 0$ (see Tardella (2010)).

Now, consider adding restrictions to the parameter vector and the sizes of the sets to explicitly obtain a specific corner solutions.

Assumption 1. Let M be a positive integer strictly greater than 1. Assume that $N = L = M$ and $\mu_i = \nu_j$ for all $1 \leq i, j \leq M$.

Assumption 1 ensures that each school reaches full capacity with individuals from the same group.

Assumption 2. For each $1 \leq i \leq N$, suppose there exists $1 \leq \zeta_i \leq L$ such that $c_{i\,\zeta_i} < c_{i\,j}$ for all $1 \leq j \leq L$ with $j \neq \zeta_i$. Furthermore, assume that $\zeta_i \neq \zeta_j$ for all $1 \leq i, j \leq L$ with $i \neq j$.

Assumption 2 imposes that each individual is optimally matched with their top choice school, ensuring a distinct best fit for each individual. Note that Assumptions 1 and 2 imply immediately that the solution to the linear model is:

$$\pi^* = [\pi^*_{ij}] = \begin{cases} \mu_i & \text{if } j = \zeta_i, \\ 0 & \text{otherwise.} \end{cases} \tag{12}$$

In fact, for any other matching $\pi \in \Pi(\mu, \nu)$,

$$C(\pi, \theta) = \sum_{i=1}^{N} \sum_{j=1}^{L} d_{ij} + c_{ij}\pi_{ij} > \sum_{i=1}^{N} \sum_{j=1}^{L} d_{ij} + \sum_{i=1}^{N} c_{i\,\zeta_i} \sum_{j=1}^{L} \pi_{ij} = C(\pi^*, \theta).$$

Assumption 3. Let $\widetilde{c_i} = \min_{\substack{1 \leq j \leq L \\ j \neq \zeta_i}} \{c_{ij}\}$ satisfy $\widetilde{c_i} > c_{i\,\zeta_i} + a_{i\,\zeta_i}\mu_i^2(1 - 1/L)$ for $1 \leq i \leq N$.

Assumption 3 tells us that preferences must be such that *the top choice* only based on c_{ij} is at least $a_{i\,\zeta_i}\mu_i^2(1 - 1/L)$ better than the other ones. By combining Assumptions 1, 2 and 3 we show that the solution to $\mathcal{P}_1$, in the integer setting, is given by (12).

Theorem 4.3. *Under Assumptions 1, 2 and 3, the optimal matching for the quadratic model in the integer setting is* (12).

Proof. Let π be an arbitrary matching different from π^*. Then,

$$C(\pi; \theta) = \sum_{i=1}^{N} \sum_{j=1}^{L} d_{ij} + c_{ij}\pi_{ij} + a_{ij}\pi_{ij}^2 \geq \sum_{i=1}^{N} \sum_{j=1}^{L} d_{ij} + \sum_{i=1}^{N} \left(\sum_{j=1}^{L} c_{ij}\pi_{ij} + a_{i\,\zeta_i} \sum_{j=1}^{L} \pi_{ij}^2 \right).$$

Now, consider i such that $\pi_{i\,\zeta_i} < \mu_i$. Due to the integer nature of π, $\pi_{i\,\zeta_i} \leq \mu_i - 1$. Hence

$$\begin{aligned}
\sum_{j=1}^{L} c_{ij}\pi_{ij} &= c_{i\,\zeta_i}\pi_{i\,\zeta_i} + \sum_{j\neq\zeta_i} c_{ij}\pi_{ij} \\
&\geq c_{i\,\zeta_i}\pi_{i\,\zeta_i} + \widetilde{c}_i(\mu_i - \pi_{i\,\zeta_i}) \\
&= \widetilde{c}_i\mu_i - \pi_{i\,\zeta_i}(\widetilde{c}_i - c_{i\,\zeta_i}) \\
&\geq \widetilde{c}_i\mu_i - (\mu_i - 1)(\widetilde{c}_i - c_{i\,\zeta_i}) \\
&= \mu_i c_{i\,\zeta_i} + \widetilde{c}_i - c_{i\,\zeta_i}.
\end{aligned}$$

On the other hand, consider the function $f : \mathbb{R}^{L-1} \to \mathbb{R}$ defined by

$$f(x_1, \ldots, x_{L-1}) = x_1^2 + \cdots + x_{L-1}^2 + (\mu_i - x_1 - \cdots - x_{L-1})^2.$$

Note that the set $x_j^* = \mu_i/L$ minimizes f. As a consequence,

$$\sum_{j=1}^{L} \pi_{ij}^2 = f(\pi_{i1}, \ldots, \pi_{i\;L-1}) \geq \sum_{j=1}^{L} \left(\frac{\mu_i}{L}\right)^2 = \frac{\mu_i^2}{L}.$$

Combining these results, we have

$$C(\pi;\theta) \geq \sum_{i=1}^{N}\sum_{j=1}^{L} d_{ij} + \sum_{i=1}^{N} \mu_i c_{i\,\zeta_i} + \widetilde{c}_i - c_{i\,\zeta_i} + a_{i\,\zeta_i}\left(\frac{\mu_i^2}{L}\right) > C(\pi^*;\theta). \qquad \blacksquare$$

Example 4.4. The following examples were computed using Mathematica 14.1. For each case, we present the parameter matrices d, c, and a (where applicable), along with the optimal matching matrix π^*, obtained using the appropriate optimization method.

For the linear model, with $N = L = 4$ and $\mu_i = \nu_j = 50$, the optimal matching was computed using `LinearOptimization`:

$$d = \begin{bmatrix} 32 & 83 & 82 & 37 \\ 47 & 75 & 56 & 45 \\ 87 & 74 & 79 & 4 \\ 40 & 55 & 94 & 14 \end{bmatrix}, \quad c = \begin{bmatrix} 76 & 77 & 83 & 6 \\ 74 & 98 & 7 & 41 \\ 6 & 86 & 8 & 70 \\ 88 & 17 & 40 & 96 \end{bmatrix}, \quad \pi^* = \begin{bmatrix} 0 & 0 & 0 & 50 \\ 0 & 0 & 50 & 0 \\ 50 & 0 & 0 & 0 \\ 0 & 50 & 0 & 0 \end{bmatrix}.$$

For the quadratic model, with $N = L = 4$ and $\mu_i = \nu_j = 20$,

$$d = \begin{bmatrix} 88 & 88 & 100 & 91 \\ 19 & 42 & 37 & 69 \\ 81 & 87 & 9 & 50 \\ 66 & 18 & 77 & 91 \end{bmatrix}, \quad c = \begin{bmatrix} 989 & 24 & 975 & 941 \\ 673 & 612 & 684 & 9 \\ 20 & 352 & 387 & 380 \\ 675 & 687 & 44 & 697 \end{bmatrix}, \quad a = \begin{bmatrix} 9 & 3 & 8 & 9 \\ 6 & 8 & 3 & 2 \\ 1 & 7 & 8 & 3 \\ 9 & 5 & 2 & 6 \end{bmatrix},$$

the optimal matching, obtained using `QuadraticOptimization`, is

$$\pi^* = \begin{bmatrix} 0 & 20 & 0 & 0 \\ 0 & 0 & 0 & 20 \\ 20 & 0 & 0 & 0 \\ 0 & 0 & 20 & 0 \end{bmatrix},$$

Hence, the result is in accordance with Theorem 4.3

Examples 4.1 and 4.2 demonstrate that the solution to $\mathcal{P}_Q$ can be either interior or a corner solution, unlike the classical linear model. However, under the assumptions of Theorem 4.3, the solution is always a corner solution, as illustrated in Example 4.4. Notably, our model introduces a novel approach by incorporating heterogeneous quadratic costs, a departure from standard quadratic regularization models, which typically assign a uniform value to a_{ij} and serve a different objective. This heterogeneity allows the model to capture the effects of congestion costs, which distort matchings that would otherwise be optimal based on preferences or affinity. As a result, assignments that would naturally occur in an unconstrained setting may not materialize due to the increasing marginal costs induced by congestion. Solving $\mathcal{P}_Q$ can be accomplished through numerical quadratic convex optimization. However, obtaining an analytical solution remains highly complex, as the optimal solution may or may not be a corner solution.

Having explored the specific cases where the solution is either a corner or interior solution, we now turn to the general case for $N = L = 2$, disregarding any assumption. The following calculations were obtained using Mathematica 14.1. By solving (10) and (11), we identified four parametric solution families that require $\mu_1 + \mu_2 = \nu_1 + \nu_2$. Three of these families are discarded because they correspond to degenerate cases: the first case holds when $a_{12} + a_{22} = 0$, the second case holds when $a_{11} + a_{12} + a_{21} + a_{22} = 0$ and $\mu_2 = (2a_{12}(\nu_1 + \nu_2) + 2\nu_1(a_{21} + a_{22}) - c_{11} + c_{12} + c_{21} - c_{22})/(2a_{12} + 2a_{22})$ and the third case holds when $a_{12} + a_{22} = 0$, $a_{11} + a_{21} = 0$ and $\nu_1 = (2\nu_2 a_{22} + c_{11} - c_{12} - c_{21} + c_{22})/(2a_{21})$. These unfeasible conditions leave us with one valid solution family, given by $\xi_2^* = \xi_1^* + (2(a_{11}a_{12} + a_{12}a_{21} + a_{11}a_{22} + a_{21}a_{22})\mu_2 - 2(a_{11}a_{12} + a_{11}a_{22})\nu_1 - 2(a_{11}a_{12} + a_{12}a_{21})\nu_2 + (a_{12} + a_{22})(c_{21} - c_{11}) + (a_{11} + a_{21})(c_{22} - c_{12}))/(a_{11} + a_{12} + a_{21} + a_{22})$, $\lambda_1^* = (-\xi_1^* a_{21} - \xi_2^*(a_{12} + a_{21} + a_{22}) + 2(a_{12}a_{21} + a_{21}a_{22})\mu_2 - 2a_{12}a_{21}\nu_2 + a_{22}c_{21} + a_{21}c_{22} - a_{21}c_{12} - a_{12}c_{21})/(a_{12} + a_{22})$ and $\lambda_2^* = (-\xi_1^* a_{22} - \xi_2^* a_{12} - 2a_{12}a_{22}\nu_2 - a_{22}c_{12} - a_{12}c_{22})/(a_{12} + a_{22})$ where ξ_1^* is free. By plugging these equalities into (9), we obtain the optimal matching when all the resulting expressions are strictly greater than zero. A detailed analysis to guarantee that $\pi_{ij}^* > 0$ was performed by reducing inequalities programmatically, but the numerous inequalities generated are omitted here. This analysis establishes a well-defined parameter space where the solution remains interior.

Given the specific cases analyzed above, it becomes evident that there is little hope of determining analytically whether solutions are interior or corner as N and L increase beyond 2. While the examples for $N = L = 2$ allowed us to identify some conditions under which solutions are either interior or corner, as the dimension of the problem grows, these conditions become increasingly complex and indeterminate.

The case $N = L$ becomes particularly relevant when considering the healthcare sector, where

certain hospital networks are designated for specific types of diseases. This also applies to health insurance systems, such as SIS, Essalud, and EPS. Let us see this in detail in Section 5.

5 Applications

The formulation in Problem $\mathcal{P}_1$ is particularly relevant in contexts where congestion costs significantly affect the allocation of resources. Unlike models with linear costs, the quadratic cost structure accounts for congestion effects indirectly by making overburdened facilities increasingly costly. This feature is crucial in understanding inefficiencies in the Peruvian healthcare and education sectors, where access is heavily determined by proximity to schools and bureaucratic efficiency in medical centers.

5.1 Healthcare: The Impact of Bureaucratic and Geographic Congestion

In the Peruvian healthcare system, individuals are theoretically assigned to facilities based on their insurance type—whether EsSalud (public insurance for formal workers), SIS (universal public insurance for low-income individuals), or private insurance (EPS), and, more specifically, they are assigned based on their illness. However, the system suffers from severe inefficiencies and inadequate infrastructure, with 76% of facilities lacking sufficient capacity to provide proper care (Defensoría Pueblo, 2020), which leads, among other factors, to a poor allocation.

These inefficiencies are further illustrated by alarming statistics (Velásquez, 2020). Peru recorded the highest per capita COVID-19 mortality rate globally (Anaya-Montes and Gravelle, 2024), highlighting the system's fragility. The shortage of medical personnel remains critical, with only 4 doctors per 10,000 inhabitants, far below the WHO-recommended threshold of 43 (Infobae Médicos, 2024). The deficit extends to hospital infrastructure, with only 1.6 hospital beds per 1,000 people, well below regional averages (Banco Mundial, 2023). Inefficient patient referral systems and high deferral rates further increase waiting times and restrict access to essential services (Huerta-Rosario et al., 2019; Power BI Healthcare Data, 2024a,b).

Geographic constraints further reinforce these inefficiencies. Traffic congestion in Lima[8] and other major cities significantly increases travel costs, deterring patients from seeking care at facilities that may have better capacity but require longer commutes. This behavior is well captured in our model by the term $\sum_{i,j} a_{ij}\pi_{ij}^2$. It is important to note that congestion is not limited to geographic factors. Bureaucratic hurdles often prevent patients from receiving specialized care at appropriate institutions (Power BI Healthcare Data, 2024a,b). Thus, it is necessary to consider a quadratic structure to model the fact that if many individuals of the same type seek care at the same hospital, being processed by the same subsystem within the medical center leads to saturation and congestion. We illustrate this in the following examples.

Example 5.1. In this example, we aim to represent the healthcare sector scenario, where three groups of patients are theoretically assigned to a specific type of medical center: SIS, EsSalud,

[8]The World Bank estimates that traffic congestion alone costs Peru 1.8% of its GDP annually. Similar economic losses have been reported in major cities such as Mumbai, São Paulo, and Jakarta (Kikuchi and Hayashi, 2020) due to excessive congestion.

or EPS. However, due to quadratic costs, patient distribution is not strictly confined to these categories, leading to cross-assignments between systems. Given the following parameters:

$$a = \begin{bmatrix} 2.0 & 1.0 & 2.0 \\ 1.0 & 2.0 & 2.0 \\ 2.0 & 1.0 & 2.0 \end{bmatrix}, \; c = \begin{bmatrix} 1.0 & 10.0 & 10.0 \\ 10.0 & 1.0 & 10.0 \\ 10.0 & 10.0 & 1.0 \end{bmatrix}, \; d = I_{3\times 3}$$

and

$$\mu = \begin{bmatrix} 20.0 \\ 20.0 \\ 20.0 \end{bmatrix} = \nu.$$

The optimal solution π^* under these conditions is:

$$\pi^* = \begin{bmatrix} 6.8074 & 7.1959 & 5.9966 \\ 8.6351 & 5.6081 & 5.7567 \\ 4.5574 & 7.1959 & 8.2466 \end{bmatrix}.$$

This solution highlights the deviations from a strict one-to-one patient allocation, as the quadratic cost terms allow for cross-assignments that would not occur in a purely linear model. For comparison, when $a = 0$, meaning there are no quadratic costs, the optimal assignment is:

$$\pi^* = \begin{bmatrix} 20.0 & 0 & 0 \\ 0.0 & 20.0 & 0.0 \\ 0.0 & 0 & 20.0 \end{bmatrix}.$$

Here, patients are strictly assigned to their designated medical system, as expected in the absence of congestion effects.

Example 5.2. In this example, we analyze a scenario where the linear costs c_{ij} are identical across all assignments, meaning there are no inherent preferences between different allocation routes. However, the quadratic congestion costs a_{ij} vary, influencing the final distribution of assignments. Despite the presence of optimal routes under linear costs, congestion effects lead to deviations in the allocation. The parameters are as follows:

$$a = \begin{bmatrix} 1.0 & 2.0 & 2.0 \\ 2.0 & 1.0 & 2.0 \\ 2.0 & 2.0 & 1.0 \end{bmatrix}, \quad c = \begin{bmatrix} 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \end{bmatrix}, \quad d = \begin{bmatrix} 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \end{bmatrix}$$

$$\mu = \begin{bmatrix} 20.0 \\ 20.0 \\ 20.0 \end{bmatrix}, \quad \nu = \begin{bmatrix} 20.0 \\ 20.0 \\ 20.0 \end{bmatrix}.$$

The optimal solution π^* under these conditions is:

$$\pi^* = \begin{bmatrix} 10.0 & 5.0 & 5.0 \\ 5.0 & 10.0 & 5.0 \\ 5.0 & 5.0 & 10.0 \end{bmatrix}.$$

This result highlights the impact of congestion costs. Even though all routes have the same linear cost, the quadratic term introduces distortions in the allocation, preventing a strict adherence to any single preferred matching pattern. Instead, the system distributes assignments to mitigate excessive congestion, leading to deviations from what would be optimal under purely linear costs.

Example 5.3. In this example we compare the standard quadratic regularization model with our proposed heterogeneous congestion cost model. Both cases share the same linear costs c_{ij} and distance factors d_{ij}, as well as the same supply and demand constraints:

$$c = \begin{bmatrix} 1.0 & 5.0 & 5.0 \\ 5.0 & 1.0 & 5.0 \\ 5.0 & 5.0 & 1.0 \end{bmatrix}, \quad d = \begin{bmatrix} 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \end{bmatrix}$$

$$\mu = \begin{bmatrix} 20.0 \\ 20.0 \\ 20.0 \end{bmatrix}, \quad \nu = \begin{bmatrix} 20.0 \\ 20.0 \\ 20.0 \end{bmatrix}.$$

In the standard quadratic regularization model, a_{ij} is uniform:

$$a = \begin{bmatrix} 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \end{bmatrix}$$

yielding the optimal allocation:

$$\pi^* = \begin{bmatrix} 8.0 & 6.0 & 6.0 \\ 6.0 & 8.0 & 6.0 \\ 6.0 & 6.0 & 8.0 \end{bmatrix}.$$

In contrast, our model introduces heterogeneity in congestion costs:

$$a = \begin{bmatrix} 2.0 & 1.0 & 1.0 \\ 1.0 & 2.0 & 1.0 \\ 1.0 & 1.0 & 2.0 \end{bmatrix}$$

leading to a different optimal allocation:

$$\pi^* = \begin{bmatrix} 4.8 & 7.6 & 7.6 \\ 7.6 & 4.8 & 7.6 \\ 7.6 & 7.6 & 4.8 \end{bmatrix}.$$

Unlike the uniform model, this formulation better captures congestion differences, reducing allocations where costs are higher and redistributing demand accordingly. This results in a more realistic representation of congestion-driven inefficiencies.

It is worth mentioning that the model is highly flexible, allowing us to analyze additional cases. For instance, instead of considering the matching between three groups of patients and the three main healthcare networks in Peru, we could group patients by type of illness and medical centers by their specialization. The existence of delays and long queues reveals frictions in the matching process, further supporting the applicability of our model.

5.2 Education: Congestion Costs and School Choice Constraints

The Peruvian education system is highly complex and decentralized, unlike centralized models in countries such as China, South Korea, and France. This decentralization has resulted in significant heterogeneity in educational quality, particularly between urban and rural areas. Unlike France, where an efficient transport network helps mitigate congestion-related issues in school assignments (Eurydice - European Commission, 2024), Peru's fragmented structure exacerbates disparities in access to education, infrastructure, and resources.

Despite this decentralization, our model remains relevant for understanding key educational dynamics and offers valuable insights if parts of the system, or even specific subsystems such as the High-Performance Schools (COAR), become more centralized. Indeed, as highlighted by (Alba-Vivar, 2025) in line with (Agarwal and Somaini, 2019), transportation in Lima plays a crucial role in educational access. A 17% reduction in travel time (equivalent to 30 minutes per day) increased enrollment rates by 6.3%, underscoring the importance of mobility constraints in shaping educational outcomes.

Moreover, Peru is characterized by severe congestion along major thoroughfares (World Bank, 2024; IFSA-Butler, 2024). As more individuals travel along the same routes, congestion intensifies, making it essential to incorporate congestion costs into the model. This effect cannot be captured by a linear structure, particularly when individuals are clustered by geographic location.

Additionally, stronger geographic constraints, such as those in the Andes and the Amazon, create highly congested access routes, including narrow bridges over rivers and limited transportation corridors. These natural barriers further justify the introduction of a quadratic term to account for congestion effects.

The following examples illustrate the impact of congestion costs in the proposed model.

Example 5.4. This example illustrates how introducing heterogeneous quadratic costs $a_{ij}\pi_{ij}^2$ distorts student allocation compared to a purely linear preference-based model. In many developed countries, such as France or Switzerland, well-developed metro systems allow students to access top schools regardless of distance. However, in Peru, inadequate public transportation significantly affects school choice, leading to inefficient assignments. We consider three groups of students and three types of schools, where c_{ij} represents student preferences, including perceived school quality and distance constraints. Without congestion costs, students would be perfectly

sorted into their most preferred schools. The parameters are as follows:

$$a = \begin{bmatrix} 4.0 & 2.0 & 3.0 \\ 4.0 & 2.0 & 6.0 \\ 3.0 & 4.0 & 3.0 \end{bmatrix}, \quad c = \begin{bmatrix} 1.0 & 5.0 & 100.0 \\ 10.0 & 1.0 & 50.0 \\ 100.0 & 50.0 & 1.0 \end{bmatrix}, \quad d = \begin{bmatrix} 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \end{bmatrix}$$

$$\mu = \begin{bmatrix} 40.0 \\ 40.0 \\ 40.0 \end{bmatrix}, \quad \nu = \begin{bmatrix} 40.0 \\ 40.0 \\ 40.0 \end{bmatrix}.$$

When congestion costs are included, the optimal assignment is:

$$\pi^* = \begin{bmatrix} 16.40 & 17.07 & 6.53 \\ 15.07 & 17.65 & 7.29 \\ 8.53 & 5.28 & 26.19 \end{bmatrix}.$$

Here, students are not necessarily assigned to their most preferred schools due to congestion effects. Those who would ideally attend top schools are redirected to lower-ranked institutions, as excessive demand increases quadratic congestion costs. For comparison, when congestion costs are removed ($a = 0$), the optimal assignment is:

$$\pi^* = \begin{bmatrix} 40.0 & 0.0 & 0.0 \\ 0.0 & 40.0 & 0.0 \\ 0.0 & 0.0 & 40.0 \end{bmatrix}.$$

This result aligns perfectly with the preference-based structure of c_{ij}, as all students are assigned to their most desired schools without deviation. This example highlights how transportation inefficiencies and congestion distort the school choice process. Unlike countries with high-quality metro systems, where students can attend their ideal schools regardless of distance, in Peru, traffic congestion and poor infrastructure create a situation where even high-achieving students may not access top-tier institutions. Our model captures these effects by incorporating heterogeneous quadratic costs, providing a more realistic representation of school allocation dynamics in constrained environments.

6 Conclusions

In this paper, we developed an optimal transport model with heterogeneous quadratic regularization to account for congestion effects in matching problems. Unlike classical models that assume linear transportation costs or entropy regularization, our formulation introduces increasing marginal costs, enabling the modeling of real-world inefficiencies caused by overcrowding.

From a theoretical perspective, we demonstrated that the optimization problem retains a convex structure and that the uniqueness of the optimal assignment is guaranteed. However, we showed that analytically characterizing the solutions is nontrivial, as the system of equations derived from the KKT conditions is singular. For the particular case where the number of agent

types and entities matches ($N = L$), we provided conditions under which the model produces corner solutions, meaning that each agent type is assigned to a single entity. This result is particularly relevant in sectors where specialization and supply segmentation are crucial, such as education and healthcare.

In terms of applications, our model is relevant for various allocation problems in both the public and private sectors. In education, it captures the congestion effects that arise when too many students are assigned to specific institutions, leading to infrastructure constraints and quality deterioration. In healthcare, our formulation applies to the distribution of patients across hospitals under segmented healthcare systems, such as the Peruvian case with SIS, EsSalud, and EPS, where excessive demand in certain hospitals leads to long waiting times and service inefficiencies. Additionally, the model can be extended to labor markets where firms face increasing costs when hiring additional workers with similar profiles, a phenomenon observed in industries with capacity constraints.

Our analysis of specific cases suggests that the problem admits both interior and boundary solutions, depending on the cost structure and assignment preferences. While we derived explicit conditions for certain cases, a general analytical solution remains challenging, indicating the need for numerical methods to solve the problem in more complex scenarios.

Future extensions of this work could include incorporating uncertainty in assignment costs and exploring intertemporal congestion dynamics. Moreover, advanced computational techniques, such as mixed-integer quadratic programming and nonlinear constrained optimization methods, could be used to analyze high-dimensional and more intricate cases.

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